DESIGN OF HUMAN-MAP SYSTEMS

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BACKGROUND

- Reading maps can be difficult for most people
- Trouble transforming exocentric to egocentric perspective
- Mentally rotate the map to fit to the world of the traveler

GENDER & AGE DIFFERENCES

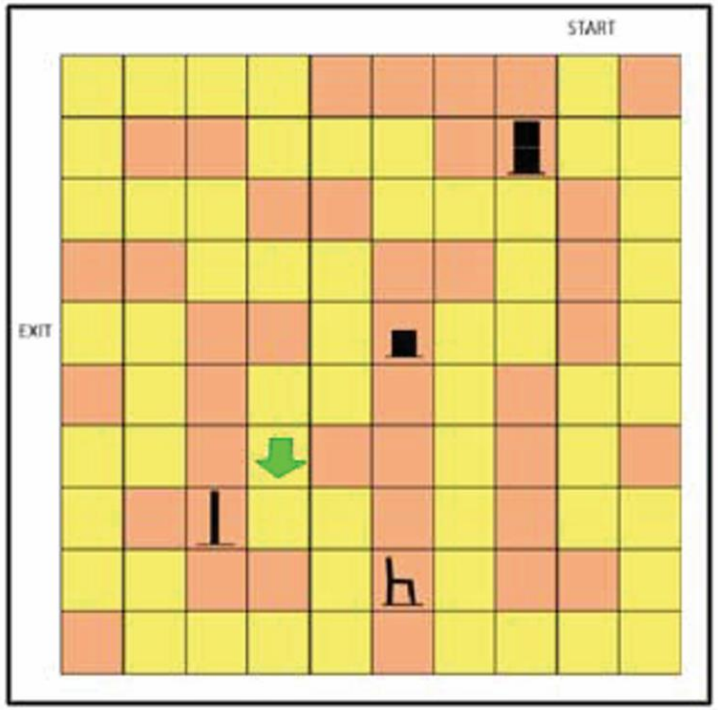
- Men can be trained to read maps more easily than women
- Females give directions that use:
  - Landmarks
  - Left/right turns
- Males give directions that use:
  - Cardinal information
  - Distance information
- Young can be trained to read maps more easily than elderly

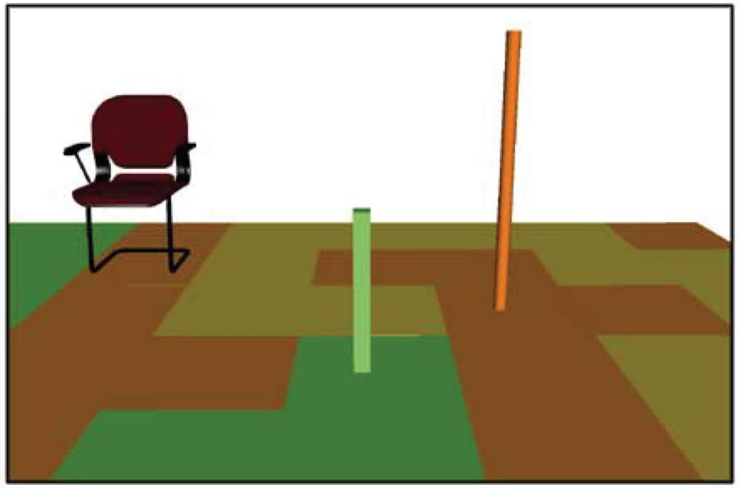
HYPOTHESIS (2007)

- Electronic egocentric map better than traditional static, electronic north-up, or head-up maps (ROUTE NAVIGATION)
  - More efficient
  - Less erroneous
  - More user-friendly
- Near egocentric perspective
  - Slanting map
  - Verbal instructions

METHOD

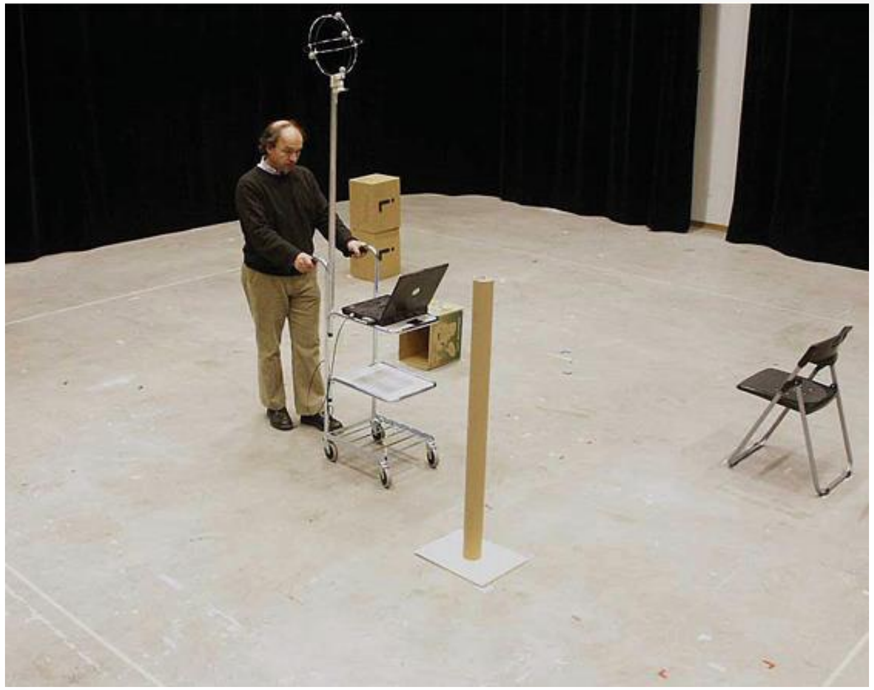
- Invisible grid of 10x10 squares in 6mx6m square
- Visible navigation aids
- Map:
  - North-up map
  - Head-up map
  - 3-D map
  - Paper map



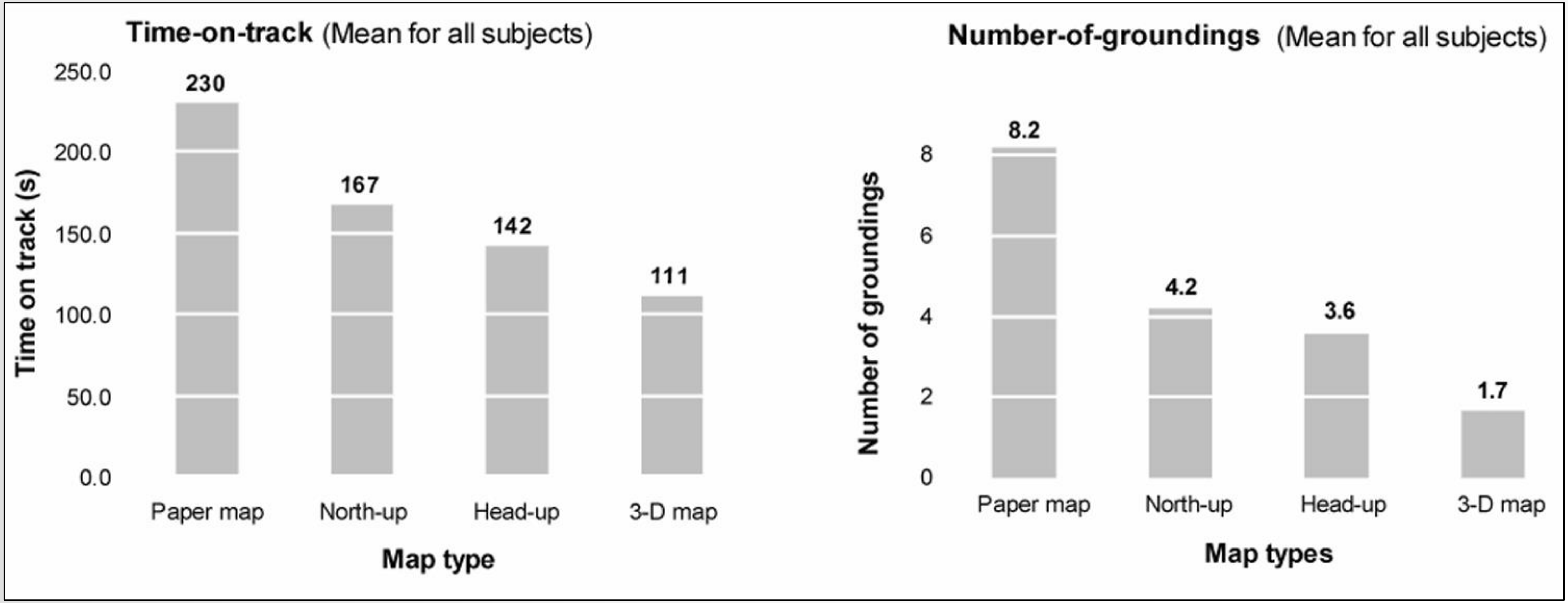


EXPERIMENT

- 45 subjects
- Drive cart through channel as fast as possible
  - Do not enter red areas of map
- Drove cart one time for each map
  - Different configurations for each



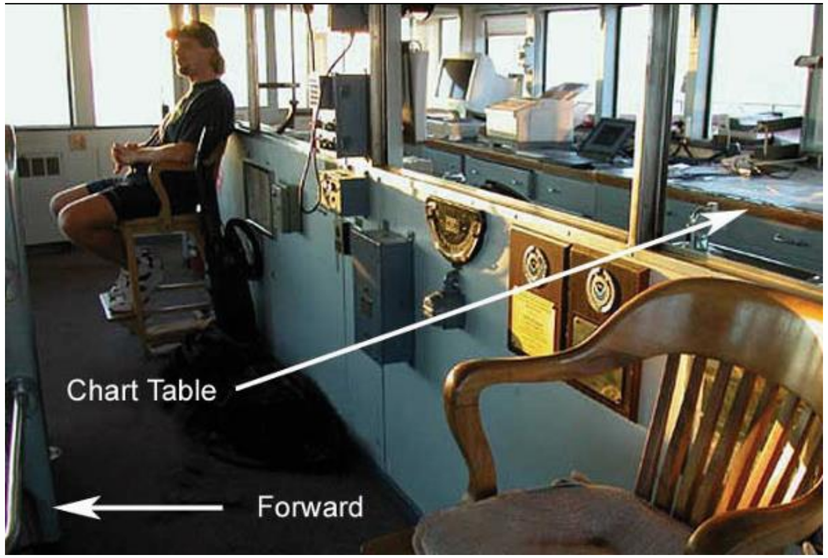
RESULTS



Time-on-track (Mean for all subjects)
Time on track (s)
250.0
200.0
150.0
100.0
50.0
0.0
230
167
142
111
Paper map
North-up
Head-up
3-D map
Map type
Number-of-groundings (Mean for all subjects)
Number of groundings
8
6
4
2
0
8.2
4.2
3.6
1.7
Paper map
North-up
Head-up
3-D map
Map types

DISCUSSION

- Egocentric maps better than exocentric
  - More efficient
  - Less erroneous
  - More user-friendly
- Route planning or judging distance not tested
- Minimizing the mental rotations is desired



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